# Week 38 Seams and cuts

Facilitator guide keyed to the revised student packets

## Mathematical purpose

Distinguish a connected surface from its boundary components, and explain a surprising cut by the seam identification. The exact objects are thin rectangles with their two short ends identified; the physical thickness of paper is ignored.

**Seam facts.** Matching end order gives an annulus A with two boundary circles. Reversing end order gives a Möbius band B with one boundary circle. Before cutting, each is one connected surface. An edge trip follows one boundary circle and stops at its first return to its marked start; it does not cross the surface or jump across the seam.

**Exact middle cuts.** Cutting the entire center circle of A produces two annuli, each with two boundary circles: two pieces and four boundary components in total. Cutting the entire center circle of B produces one annulus with two boundary circles. This result assumes the cut follows the centerline once as a closed curve; an off-center or incomplete cut is a different experiment. Cutting B one third of the way across is one closed cut that goes round twice; it leaves two linked pieces, a thin Möbius band (one boundary circle) and a longer annulus (two): three boundary circles, so only 1+1+1+1 of the four-dot pictures fails.

**Orientation.** A transverse arrow transported continuously once around the uncut centerline returns with the same direction on A and the opposite direction on B. On B it needs two centerline trips to regain its direction. This detects the nonorientability of B; it is not a count of visible twists or a statement about knotting in space.

**Four dots.** With b available boundary circles and four distinct edge dots, the possible reachability patterns are exactly the partitions of 4 into at most b nonempty groups. Unoccupied boundary circles are allowed. Connected-piece counts alone do not answer this question.

## Entry points and limits

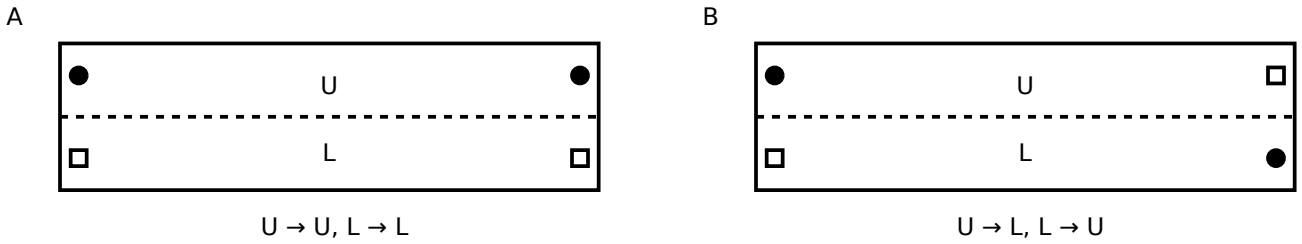
**K–1:** trace an edge through a real seam, compare dots, and observe adult cuts. **Grades 2–3:** use U/L seam connections to explain the piece count. **Grades 4–5:** separate the two component counts and explain arrow reversal. Reading can be adult-supported; arithmetic is counting to four. The abstraction is following an identification rather than the flat drawing’s apparent boundary.

**Before-use gate:** exact paper/tape bands, center cuts, removable arrows, and marker handling have not been physically pretested. K–1 must first show reliable seam tracking with an adult. All cutting is adult-only. Digital proofs and drawings do not certify a workable physical model.

## Preparation and a flexible hour

**Materials per child:** sheet 1 of the strips (week-38-materials.pdf: A, B and a spare A and B, each 260 × 45 mm), washable markers, four removable dots and the band's seven-page packet; K–1 and grades 2–3 children also get a new B from sheet 2 for Problem 7. Print sheet 1 fourteen times (eleven children, three adults) and sheet 2 three times (twelve new B), at 100% on copy paper, and check the 10 cm line. Per table: clear tape, adult scissors, and for grades 4–5 a removable transverse arrow. Grades 4–5 keep the spare A and B uncut for Problem 6.

**Prepare and physically pretest before teaching.** Cut the strips out. Write U and L on the back of each half, behind the printed letters, so both faces are marked. Join each strip dot to dot and square to square with clear tape on both faces, so the arrows line up: A with no twist, B with one half-turn. Check smooth seams without extra bridges. Trace every boundary, cut each whole centerline, and verify the expected pieces and edges with the exact materials; cut a joined new B along its dotted line until the cut closes. Check that the arrow can slide without being detached, flipped by hand, or rotated within the strip. On B the arrow comes back on the other face; that is one-sidedness, not a flip. Record what actually worked; these tests remain unperformed here.




**Shared launch, about 5 minutes.** Let children handle the bands first. Use the packet's input/process/output picture: point to the flat dot and square, join each to its mate, and locate those same marks on the band. Move a finger along a short edge segment through the seam, without completing the whole edge count. Show one start dot and the first-return stopping rule. Before P3, use the flat washer above it on page 3: P and Q share the outer trip, R lies on the inner trip. Ovals record boundary membership, not piece count or physical shape. Ask, "Where will your finger go next?"

| Time | Suggested first visit, 35–40 minutes at the tables |
|---|---|
| 0–15 min | Problems 1–2: trace every edge of A and B; place two dots. |
| 15–28 min | Problem 3, once seam tracking is reliable: the four-dot ring pictures. |
| 28–40 min | Problem 4 if there is time: predict, then an adult cuts. Share one observation. |
| Return | Problems 4–7. Grades 4–5 can take Problems 4–5, then 6–7 on another visit. |

**Readiness and pacing.** Let a child show an edge trip without reading or writing. If the child repeatedly hops across the band or cannot follow a joined seam, stay with adult-guided tracing of A; do not require the abstract four-dot patterns. For older children, do not spend the proof time finishing every picture. Keep scissors with an adult, all bands on the table, and loops away from bodies.

## K–1 answer key

**Problem 1.** A has two closed edges; B has one. Give A's two edges different colors. One color traces all of B's boundary, although the flat recipe shows it as upper and lower long edges. The joined short ends are seams, not additional exposed edges.

**Problem 2.** On A, put both dots on one boundary for a shared trip; put one on each boundary for no shared trip. On B, any two edge dots can be reached by one boundary trip. The "no one trip reaches both" request is impossible on B.

**Problem 3.** The five pictures are, from top to bottom, 4; 3+1; 2+2; 2+1+1; 1+1+1+1. The groups are boundary circles containing dots, not physical pieces.

| Picture | A before cutting | B before cutting |
|---|---|---|
| 4 | Yes | Yes |
| 3+1 | Yes | No |
| 2+2 | Yes | No |
| 2+1+1 | No | No |
| 1+1+1+1 | No | No |

**Problem 4.** Accept every prediction as a conjecture before the adult cuts. The intended exact result is two separate annular pieces from A and one longer annular piece from B. A drawing need not look like a perfect ring; preserve connectedness. If the physical result disagrees, inspect the seam and cut instead of calling the child's observation wrong.

**Problem 5.** Every resulting piece has two boundary circles. Yes, two dots can be on the same piece but on different closed edges in both cut results. The cut B is one piece with two boundaries, so "one piece" does not mean "one edge trip."

**Problem 6.** Cut A, using both pieces together, supports all five pictures. Cut B supports only 4, 3+1, and 2+2. See the complete shared table on the next page.

**Problem 7.** The cut goes round the new B twice before it closes. It leaves two linked pieces: a thin Möbius band from the middle third, with one closed edge, and a longer band with two closed edges, one of them B's old edge. Three closed edges in all, so 4, 3+1, 2+2 and 2+1+1 are yes and 1+1+1+1 is no: four dots each alone need four edges. Let children find the three edges by tracing before they answer.

**Optional support.** Place a start sticker and let the child hold it while the adult guides a finger. Say "same piece?" and "same edge trip?" as separate questions. The child can place dots and replay rather than write answers.

## Grades 2–3 answer key

**Problems 1–2.** A has two boundary circles; B has one. A permits both a pair on the same boundary and a pair on different boundaries. B permits only the first request. A complete edge trip returns to its start after following one circle, not merely after reaching a seam mark.

**Problem 3.** Before cutting, the top three patterns (4, 3+1, 2+2) work on A, and only the first (4) works on B. A needs at least three boundaries for 2+1+1 and four for 1+1+1+1; it has only two.

**Problem 4.** A cuts into two pieces; B cuts into one. Record the prediction separately from the result. These are connected components of the cut surface, not boundary-component counts.

**Problem 5.** On A, U joins U and L joins L, so the two half-strips become separate pieces. On B, U joins L and L joins U, so one can travel through both halves without leaving the same piece. Each piece is an annulus and has two boundary circles. Thus the answer to the same-piece/different-edge question is yes for both cut results.

**Problem 6.** Use all the pieces from one starting band as one collection. Other boundaries may remain unoccupied.

| Four-dot picture | Cut A total | Cut B total |
|---|---|---|
| 4 | Yes | Yes |
| 3+1 | Yes | Yes |
| 2+2 | Yes | Yes |
| 2+1+1 | Yes | No |
| 1+1+1+1 | Yes | No |

**Problem 7.** The cut along the dotted line goes round the new B twice before it closes. It leaves two linked pieces: the middle third, a thin Möbius band with one closed edge, and the outer thirds joined end to end, a band with two closed edges. Three closed edges: yes for 4, 3+1, 2+2 and 2+1+1; no for 1+1+1+1, which needs four. Compare Problem 6: cut A has four edges in two pieces, cut B two edges in one piece, the new B three edges in two pieces. The pieces do not decide the pictures; the edges do.

## A proof with the materials

Have children trace U’s continuation at the joined seam and then L’s. This establishes connectivity without a claim about how easy it feels to pull pieces apart. Next trace the boundaries independently. For cut B, mark an old edge and a newly cut edge: the entire old boundary is one circle and the new cut boundary is the other. The two sides of the center cut connect to form that single new circle.

## Grades 4–5 answer key

**Problems 1–2.** A’s upper long edge closes to itself, and its lower long edge closes to itself. Hence two boundary components. B’s upper long edge continues as the lower edge at the seam, then returns to the upper edge after the next seam encounter. Together they form one boundary component. Each uncut band is nevertheless one connected surface.

**Problem 3.** In printed order, A: yes, yes, yes, no, no. B: yes, no, no, no, no. Necessity: a pattern with k nonempty rings needs k different boundary components. Sufficiency: choose k available components and place the required number of distinct dots on each. Empty components are expressly allowed.

**Problem 4.** A gives two pieces, each with two boundary components (four total). B gives one piece with two boundary components (two total). An exact cut changes the surface; it is not a deformation proving the cut surface homeomorphic to the original.

**Problem 5.** The seam permutation of the half-strips is identity for A and the swap U↔L for B. Its cycles count the resulting connected pieces: (U)(L) versus (UL). A’s pieces inherit ordinary matching seams. In B, traversing the cut piece follows both original halves; the transverse direction reverses twice and returns unchanged. Each component is therefore an interval bundle over a circle with matching ends, an annulus. This is a separate argument for its two boundary components.

**Problem 6.** Use the two spare strips to rebuild uncut A and B, not the already cut objects. Keep the removable arrow tangent to the surface and transverse to its centerline, sliding continuously without voluntarily spinning it. On A, U→L returns U→L. On B it returns L→U relative to the original location. A second full trip on B restores U→L. This is monodromy of the transverse direction; turning the whole model in the room does not change the comparison at the same basepoint. On B the arrow comes back on the other face; that is one-sidedness, not a flip. With U and L on both faces, it now points from L toward U.

**Problem 7.** Cut A permits all five partitions; cut B permits only 4, 3+1, and 2+2. Two pieces of cut A supply four boundary components, whereas one piece of cut B supplies two. The partition data record which dots share a boundary, not whether dots can be joined by some path through the interior of a piece. If an explanation counts pieces, ask how many closed edges each piece has.

## Optional questions

“Why must we count the edges after counting the pieces?” “What remains true if the paper is crumpled without tearing?” “Would a different-looking spatial embedding change the seam calculation?” Keep extra half-turn experiments separate: seam parity determines the abstract band type, but spatial knotting or linking requires additional information.

## Proof notes sources and before-use record

## Why the seam calculation is enough

Represent a band by $[0,1] \times [-1,1]$. For A identify $(0,t)$ with $(1,t)$; for B identify $(0,t)$ with $(1,-t)$. The boundary is the image of $t=\pm 1$. Tracking these endpoints proves the uncut boundary counts. Cutting $t=0$ separates each fiber into its upper and lower halves, but the end identifications remain: their permutation cycles determine which half-strips remain connected.

For B the resulting cycle passes through U and L. Its two reversing seam identifications compose to the identity on a transverse interval. Equivalently, the new strip closes only after two original longitudinal passages, with no remaining transverse reversal. Thus it is an annulus. Its two boundary circles are the old boundary and the newly created boundary. This argument concerns an abstract surface, regardless of the model's extra visible twist.

The four-dot classification has exactly five cases because every partition of four positive units is 4, 3+1, 2+2, 2+1+1, or 1+1+1+1. A pattern can be realized precisely when there are at least as many boundary circles as nonempty groups. No further restriction comes from unused circles or dot order within a group.

## Sources and scope

Allen Hatcher, *Algebraic Topology*, Chapter 0, opening Möbius-band and core-circle discussion. This gives the surface context; the seam, cut, and dot proofs above are supplied explicitly for this activity. Chapter 0 PDF.

NRICH, *Bands and Bridges: Bringing Topology Back* (2011), strip-and-band activity precedent. Its published age range is 7–14; it does not validate the K–1 adaptation. NRICH article.

## Record before classroom use

Date and adult: _____
Exact paper, tape, markers, and strip dimensions: _____
Both seam traces, both complete cuts, and arrow transport checked: _____
K–1 readiness observed for participating children: _____
Adjustment needed before use: _____

**Status:** revised student packets and this adult guide are unpiloted. Algebraic seam checks and rendered-page inspection are not physical pretests. Do not announce classroom readiness until the exact-material tests and child readiness checks have been performed.

Key alignment: seven-page student packets F38-K-v4, F38-23-v4 and F38-45-v4, and the strips W38-M-v1. Guide prepared October 3, 2026; materials, launch, timetable and the Problem 6–7 keys revised October 23, 2026 from the review card.

# Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

### Use and return choices

Keep one adult at each of three tables; the adult performs the cuts while children predict and retrace. A first visit can stay with P1-2 edge trips and two-dot reachability, adding P3 only when seam tracking is reliable. On return, K-1 and grades 2-3 can use the middle cuts and retracing (P4-5), then the four-dot questions (P6-7). Grades 4-5 can separate the piece/boundary explanation (P4-5) from the transported-arrow investigation (P6) and classification (P7). Keep the spare A and B from sheet 1 uncut, and join a new B from sheet 2, before the return visit. Quick grades 2-3 and 4-5 children can go on to bonus Problem 1 (three- and four-lane cuts, week-38-bonus.pdf). The physical seam/cut/arrow test on guide p2 is still unperformed.